

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 2: Analog PLL – Phase Detector and Type-I PLL

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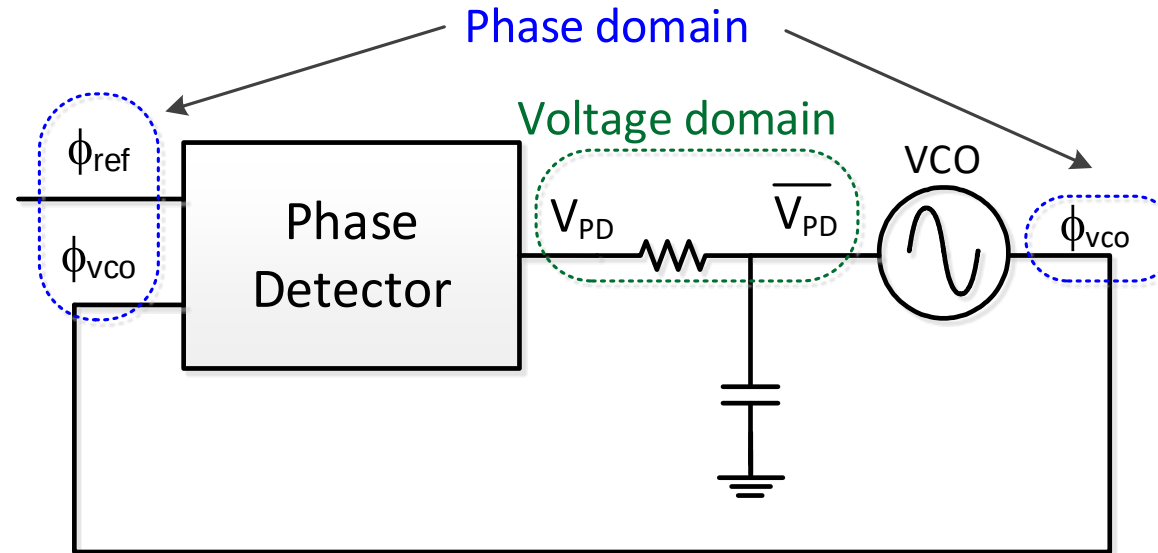


Outline

- Basic PLL building blocks
- Phase detector implementations and characteristics
 - XOR phase detector
 - Mixer phase detector
 - Sub-sampling phase detector
- Type-I PLL
 - S-domain model and loop transfer functions
 - Loop stability
 - Loop dynamics



Basic Phase Locked Loop

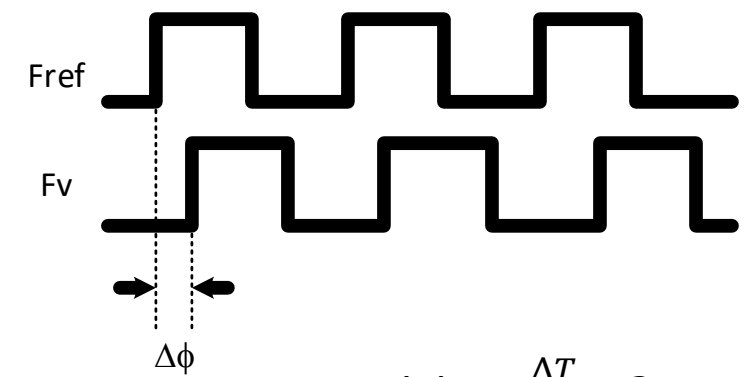
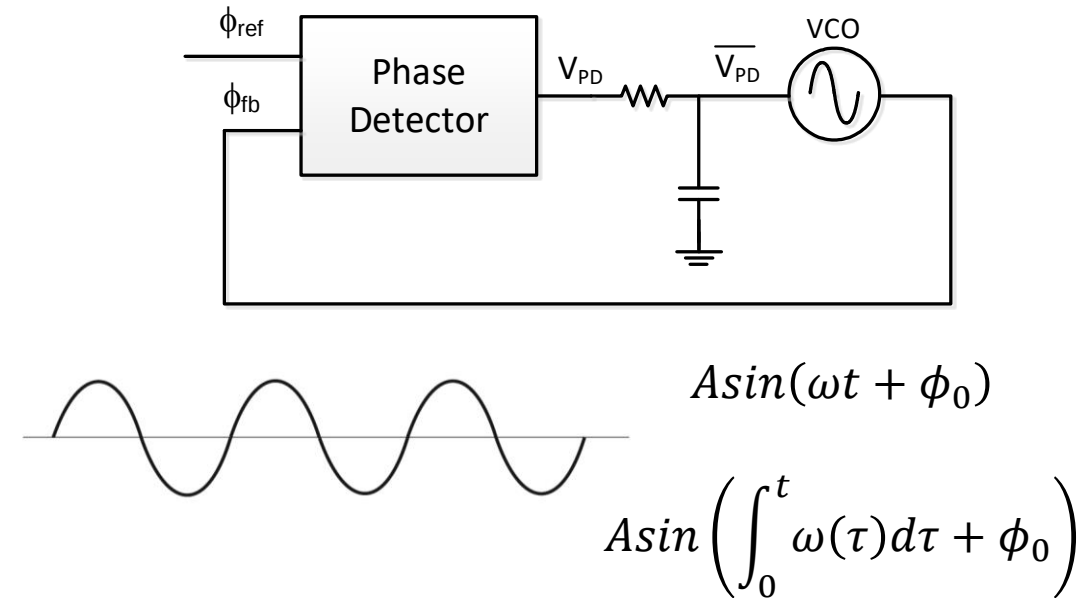
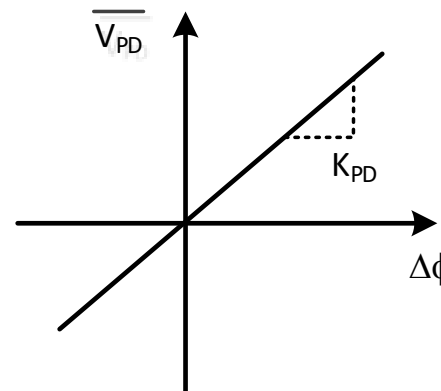


- In its simplest form, a phase locked loop consists of a phase detector, a low pass filter and a voltage-controlled oscillator.
- Phase detector (PD) generates output voltage with an average proportional to input phase difference, followed by a low pass filter.
- The filtered PD output voltage controls VCO frequency (or phase) which is fed back to PD input. For an ideal VCO, its frequency change is proportional to the input control voltage.
- The signal of interest changes from phase at PD input to voltage at PD output and back to phase at VCO output along the loop.



Phase Detector

- Phase detector (PD)
 - Inputs: two periodic inputs (reference and feedback signals)
 - Output: voltage pulses whose average value is proportional to the inputs phase difference
 - Ideal PD transfer function is linear with a slope call PD gain, K_{PD} .
 - The low pass RC filter in the following stage filters PD output V_{PD} and extracts the average voltage.



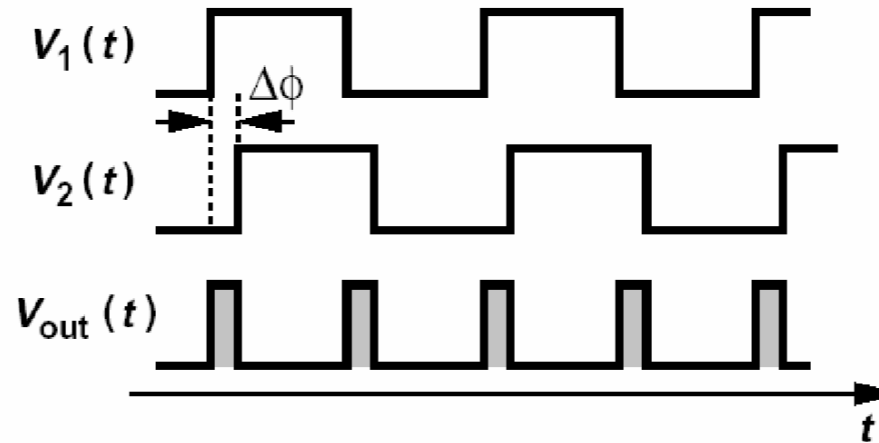
$$\Delta\phi = \frac{\Delta T}{T_{ref}} \cdot 2\pi$$



XOR Phase Detector



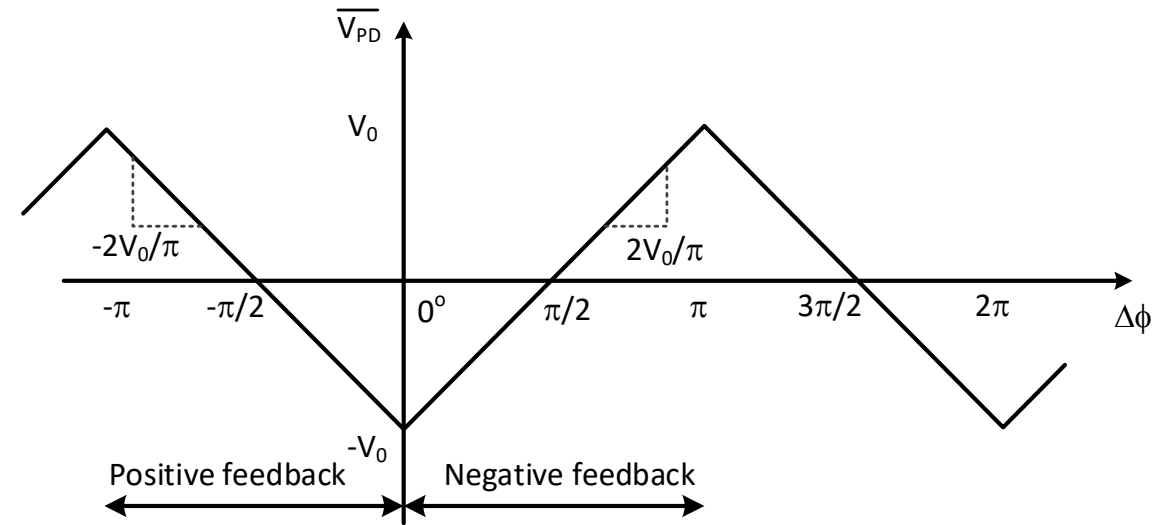
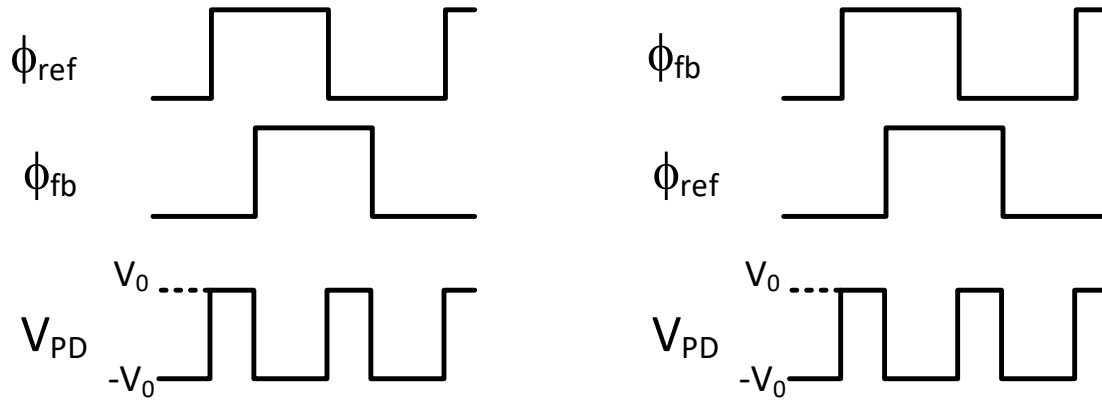
[Razavi]



- XOR (exclusive OR) gate generates pulses whose width is equal to $\Delta\phi$.
- It produces pulses at both rising and falling edges of inputs – sensitive to signal duty cycle ☹️



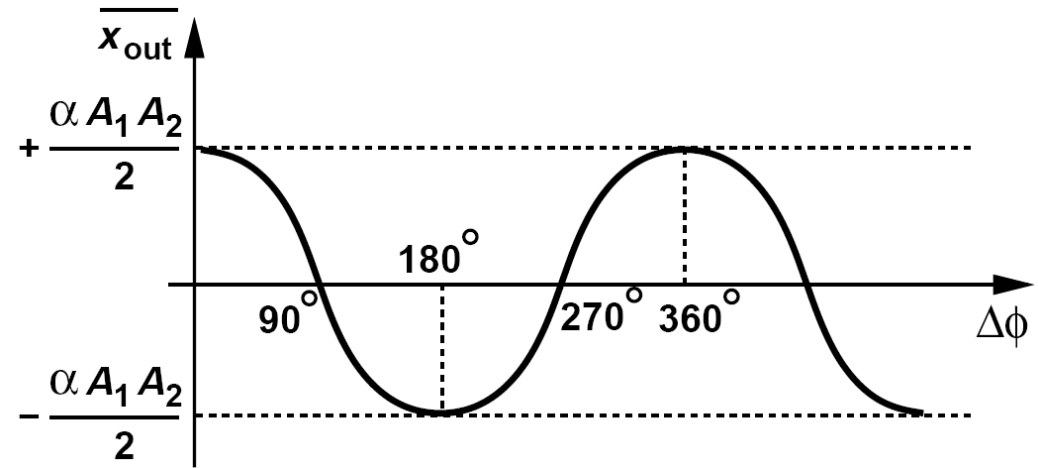
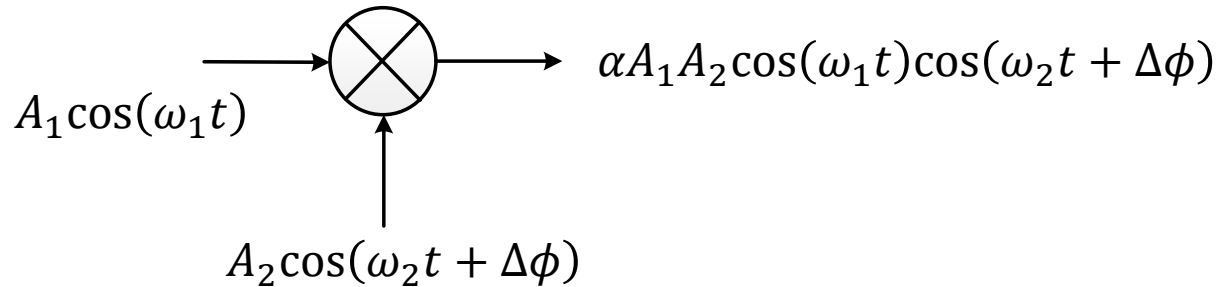
XOR Phase Detector



- XOR PD output width is the same for reference signal leading or lagging the feedback signal
 - ϕ_{ref} leading: $0 < \phi_{\text{ref}} - \phi_{\text{fb}} < \pi$, $K_{\text{PD}} = 2V_0/\pi$
 - ϕ_{ref} lagging: $-\pi < \phi_{\text{ref}} - \phi_{\text{fb}} < 0$, $K_{\text{PD}} = -2V_0/\pi$
- In the negative K_{PD} region, loop is positive feedback
- When there is frequency difference between input signals, PD can jump between different PD gain regions, PLL is not a linear system. For large frequency difference, loop may not be stable.



Mixer Phase Detector



- Mixer output: $\frac{\alpha A_1 A_2}{2} \{ \cos[(\omega_1 + \omega_2)t + \Delta\phi] + \cos[(\omega_2 - \omega_1)t + \Delta\phi] \}$
- After low pass filter $\omega_1 + \omega_2$ term is removed
- If $\omega_1 = \omega_2$, the mixer PD average output is

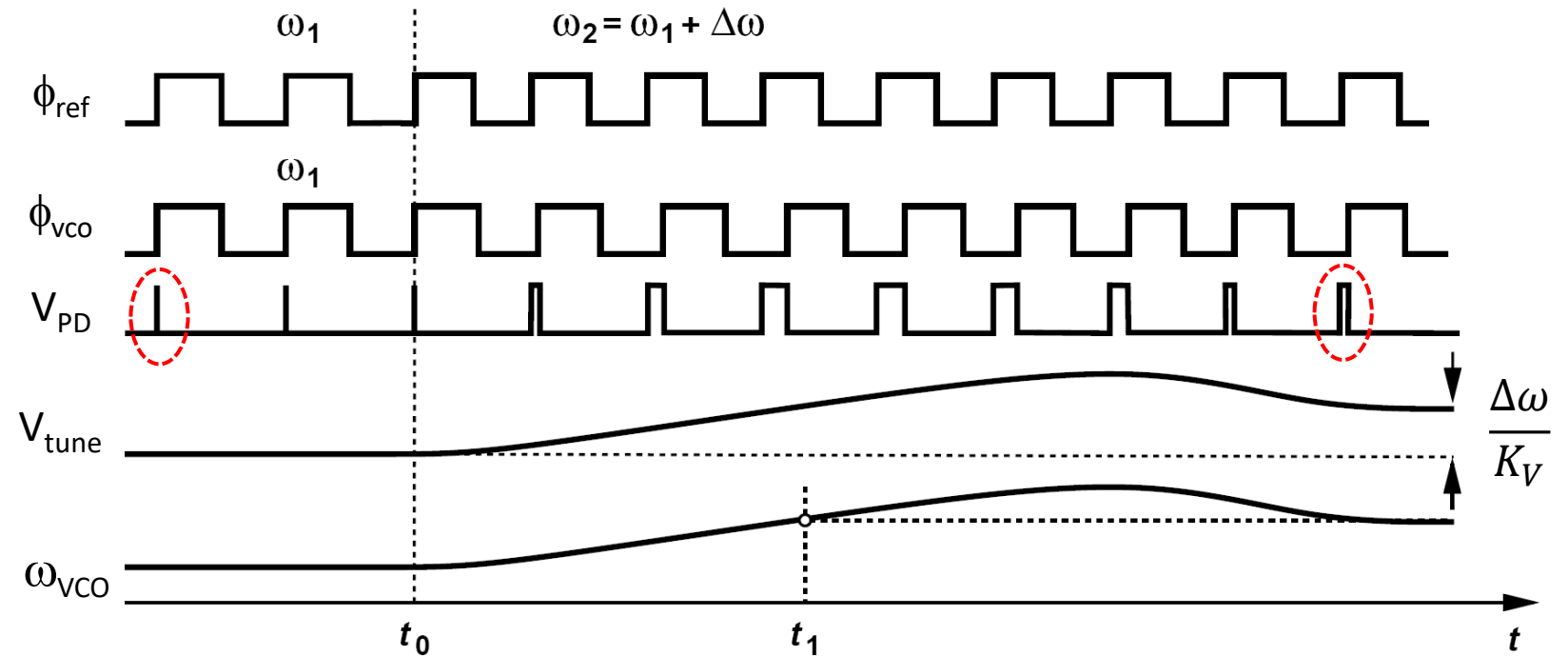
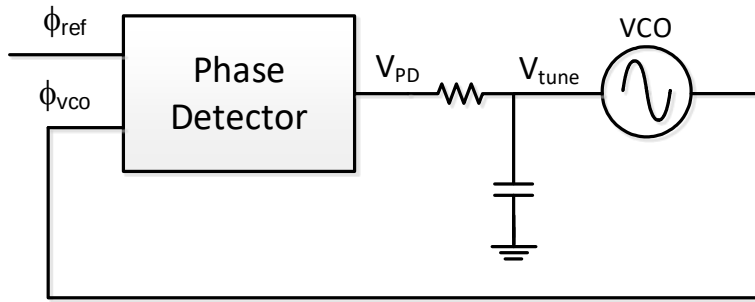
$$\overline{V_{PD}} = \frac{\alpha A_1 A_2}{2} \cos \Delta\phi$$

- PD gain $K_{PD} = \frac{d(\overline{V_{PD}})}{d\Delta\phi} = -\frac{\alpha A_1 A_2}{2} \sin \Delta\phi$; $K_{PD} \approx \frac{\alpha A_1 A_2}{2}$ for $\Delta\phi$ around 270° .
- Similar to XOR PD, have both positive and negative PD gain regions, periodic.
- Mixer PD gain is nonlinear: maximum gain at 90° , 270° , etc; minimum gain at 0° , 180° , etc.

[Razavi]



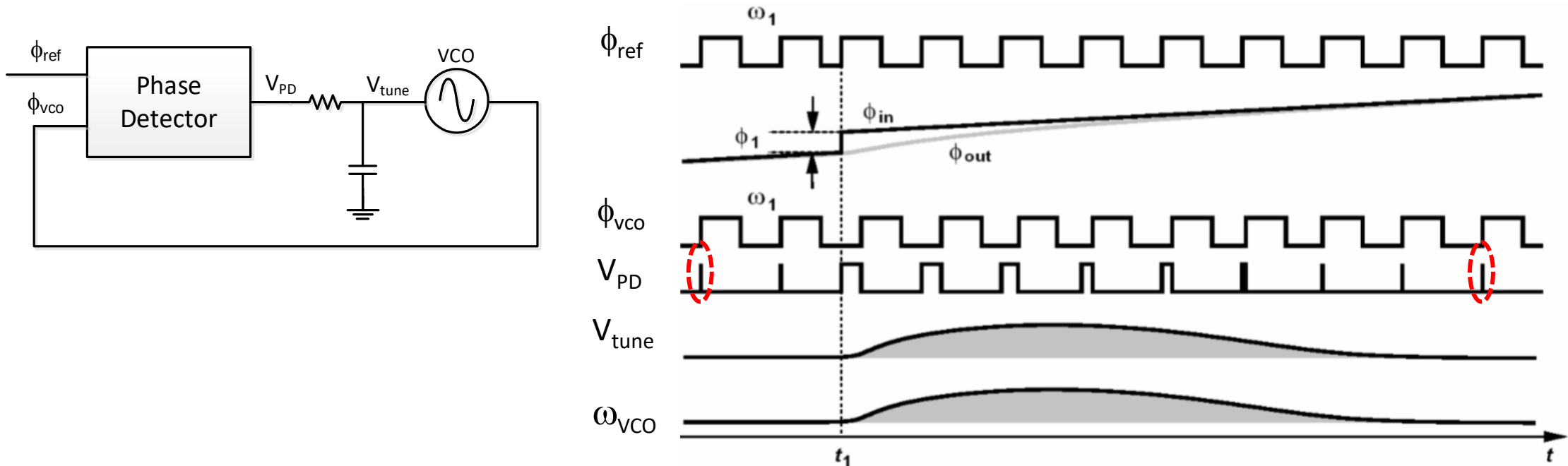
Phase Locked Loop Frequency Step Response



- In response to a frequency step at the reference input of $\Delta\omega$:
 - PD input phase difference increases and output voltage pulse width increases
 - PD output average voltage after low pass filter increases
 - VCO frequency increases that allows ϕ_{VCO} to accumulate phase faster and reduce PD input phase difference
- Lock condition: (1) $\omega_{VCO} = \omega_{ref}$ (2) $\phi_{VCO} - \phi_{ref} = \text{constant}$



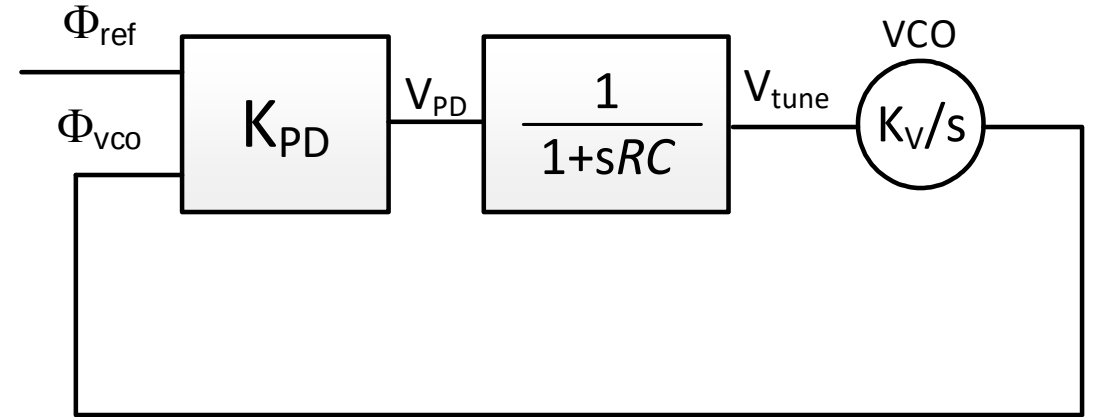
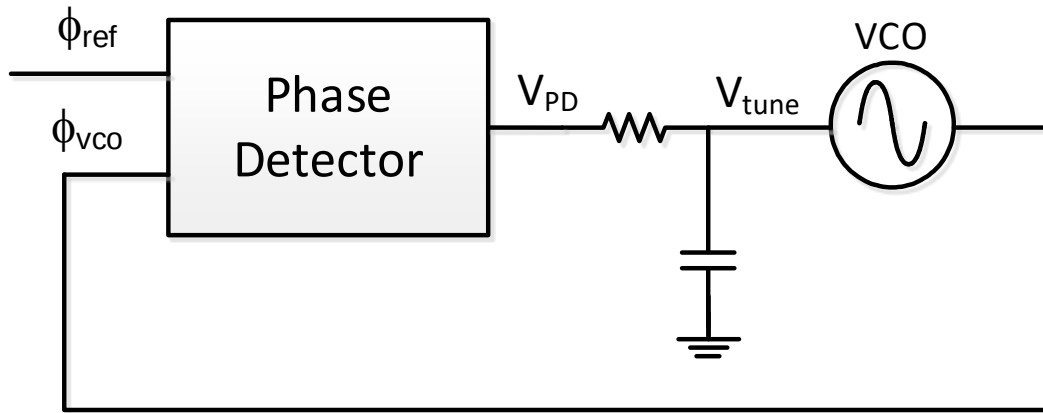
Phase Locked Loop Phase Step Response



- In response to a phase step at the reference input of $\Delta\phi$:
 - PD input phase difference increases, output voltage pulse width and average voltage V_{tune} increases
 - VCO frequency increases that allows ϕ_{vco} to accumulate phase faster and reduce PD input phase difference
 - VCO frequency returns back to ω_1 after VCO phase reaches reference phase
- Lock condition: (1) $\omega_{VCO} = \omega_{ref}$ (2) $\phi_{VCO} - \phi_{ref} = 0$



Type-I Phase Locked Loop S-domain Model



- S-domain model of building blocks

- PD: K_{PD}

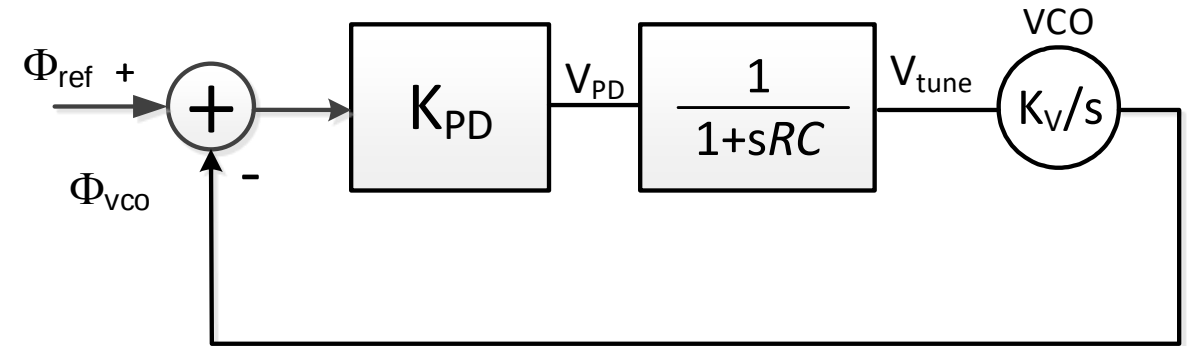
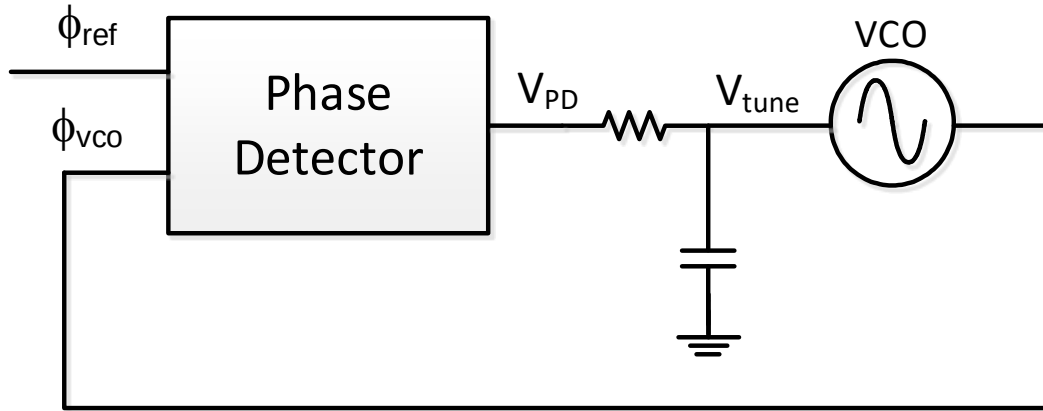
- Low pass filter: $\frac{1}{1+sRC}$

- VCO: $\frac{K_V}{s}$

- VCO characteristics: $\omega_{VCO} = \omega_0 + K_V \cdot V_{tune}$, $\phi_{VCO} = \int \omega_{VCO} dt$, $H(s)_{VCO} = \frac{K_V}{s}$



Type-I Phase Locked Loop S-domain Model



- S-domain model of open loop transfer function

$$H_{OL}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)}$$

- Closed loop transfer function

$$H_{CL}(s) = \frac{\Phi_{VCO}}{\Phi_{ref}} = \frac{H_{OL}}{1 + H_{OL}} = \frac{\frac{K_{PD} \cdot K_V}{RC}}{s^2 + \frac{s}{RC} + \frac{K_{PD} \cdot K_V}{RC}}$$

- One pole at DC in open loop transfer function: **Type-I PLL**



Loop Stability Analysis with Bode Plot

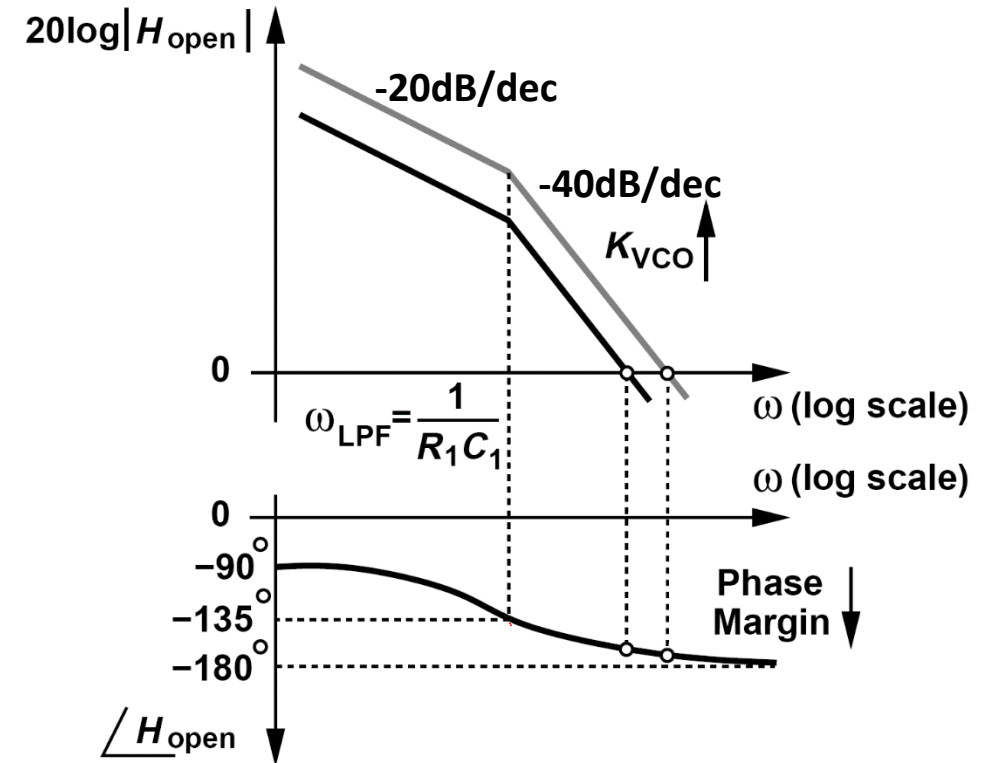
- Open loop transfer function

$$H_{OL}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)}$$

- Bode plot

→ phase margin

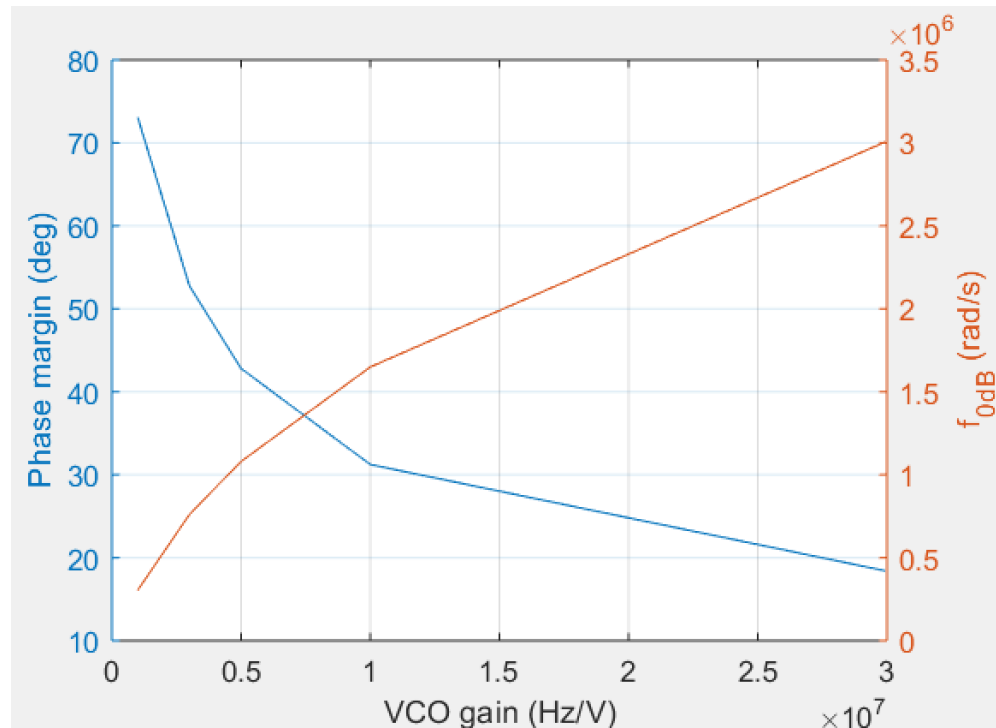
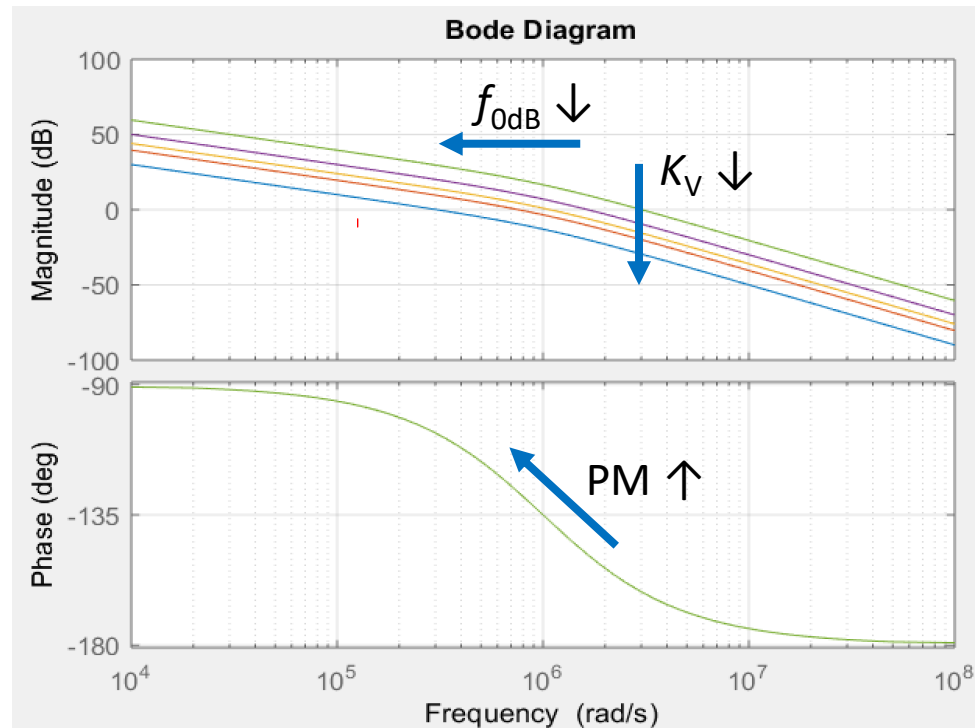
- Find the frequency where OL gain is unity, f_{0dB}
- The difference between the phase at f_{0dB} and -180° is phase margin (PM)
- Stability criterion: $PM > 0^\circ$
 - In practice, $PM > 50^\circ$
- When loop gain $\uparrow$, $f_{0dB} \uparrow$ and $PM \downarrow$



[Razavi]



Loop Stability Analysis with Bode Plot



- Matlab commands for Bode plots and phase margin analysis
 - `tf`, `bode`, `margin`



Delay Induced Phase Margin Degradation

Problem Statement

- In practice, the phase detector is often implemented using discrete time sampling
 - Phase error is sampled by PD only at each reference edge
 - S-domain model assumes a continuous time system
 - What is the condition for this assumption to be valid?
- Zero order hold (ZOH) converts a discrete-time signal to a continuous-time signal by holding each sample value for one sample interval.
 - In PLL, there is an implicit ZOH that converts the discrete phase error at PD input to the continuous time voltage at PD output
 - What is the impact of ZOH on the PLL behavior?



Delay Induced Phase Margin Degradation

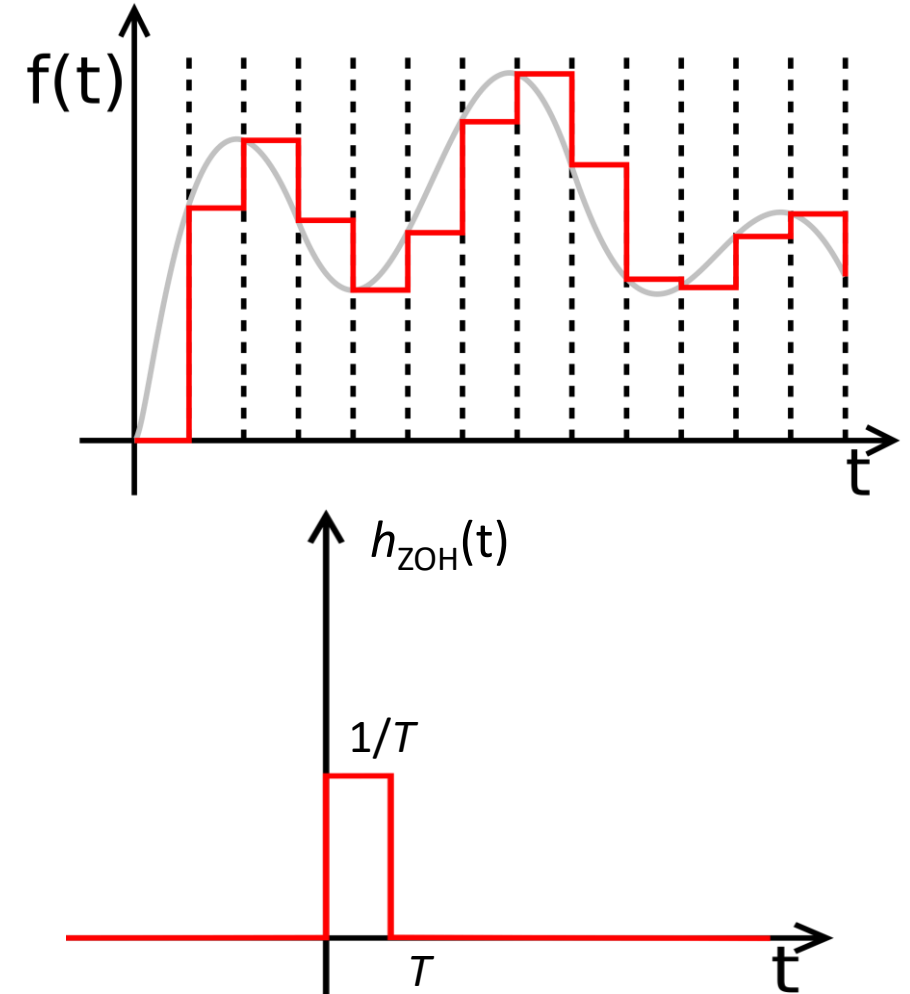
ZOH Modeling

- ZOH impulse response

- $h_{ZOH}(t) = \begin{cases} \frac{1}{T}, & 0 \leq t < T \\ 0, & \text{otherwise} \end{cases}$
- $x_{ZOH}(t) = \sum_{n=-\infty}^{\infty} x[n]h(t - nT)$

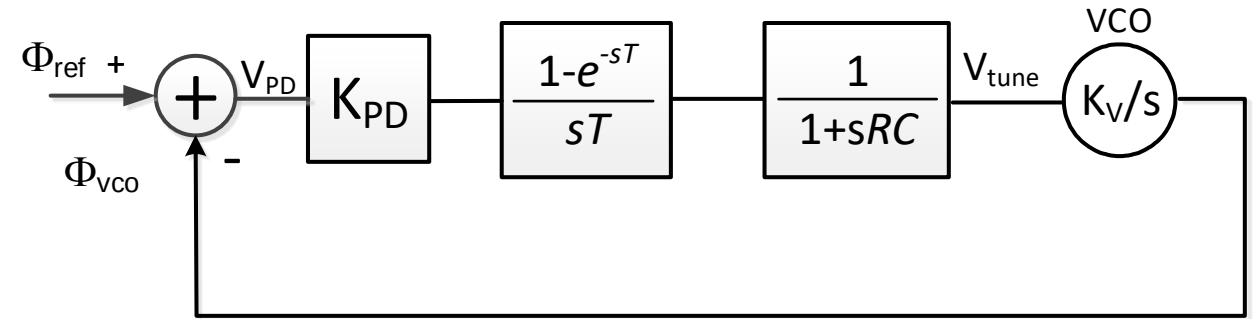
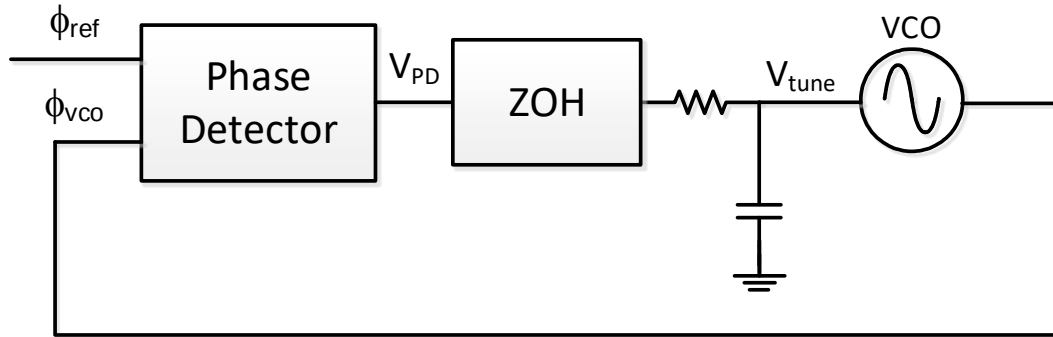
- ZOH s-domain model

- $X_{ZOH}(s) = X(s)H_{ZOH}(s)$
- $H_{ZOH}(s) = \int_0^{\infty} h_{ZOH}(t)e^{-st}dt = \frac{1-e^{-sT}}{sT}$



Delay Induced Phase Margin Degradation

A More Complete PLL Model



- S-domain model of open loop transfer function

$$H_{OL_ZOH}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)} \cdot \frac{1 - e^{-sT}}{sT} \approx \frac{K_{PD} \cdot K_V}{s(1 + sRC)} \cdot \left(1 - \frac{1}{2}sT\right)$$

- Magnitude

$$|H_{OL_ZOH}(s)| \approx |H_{OL}(s)|$$

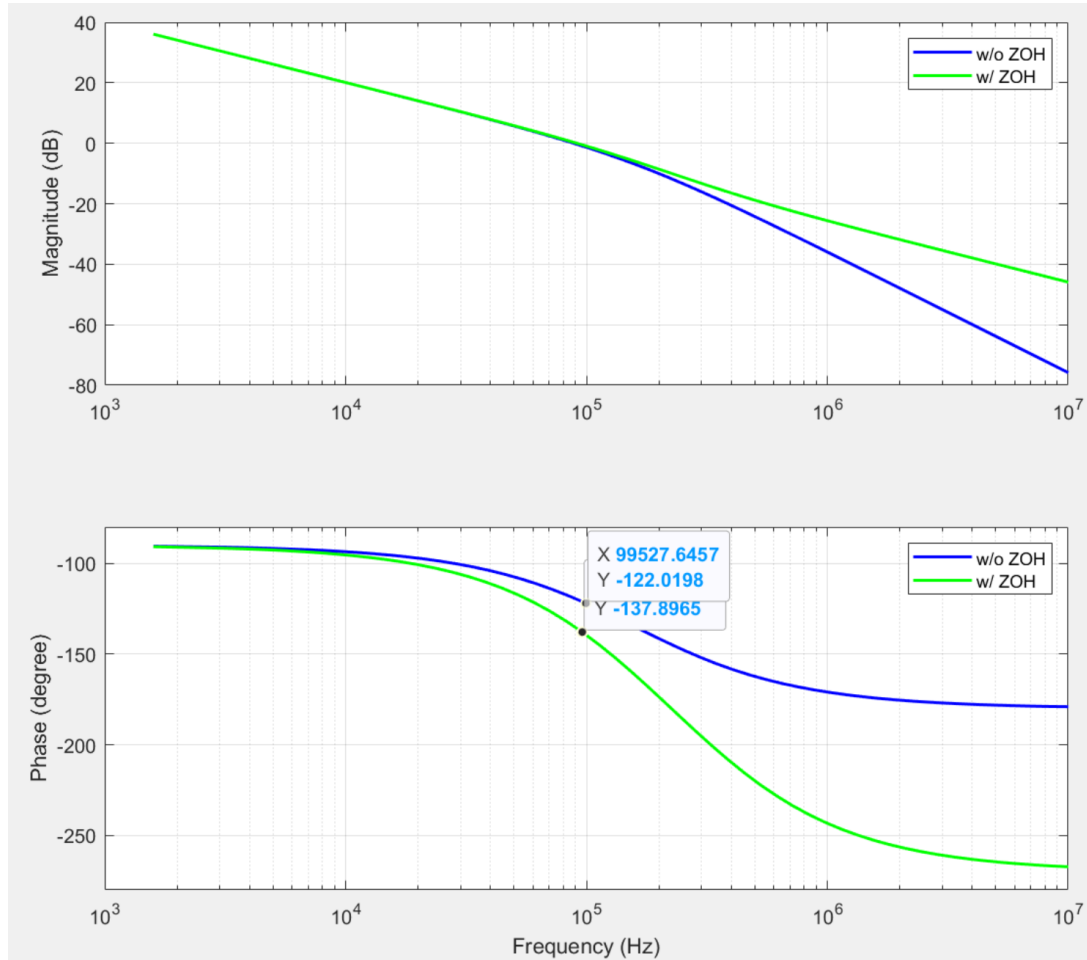
- Phase

$$\angle H_{OL_ZOH}(s) = \angle H_{OL}(s) + \angle \left(1 - \frac{1}{2}j\omega T\right) = \angle H_{OL}(s) - \arctan\left(\frac{1}{2}\omega T\right)$$



Delay Induced Phase Margin Degradation

A More Accurate Bode Plot



PM w/o ZOH 60.9°

PM w/ ZOH 44.1°

$$\angle H_{OL_ZOH}(s) = \angle H_{OL}(s) - \arctan\left(\frac{1}{2}\omega T\right)$$

- At loop unity gain frequency f_{0dB} where phase margin is calculated, phase delay from ZOH is $\frac{1}{2}\omega_{0dB}T_{ref} = \pi f_{0dB}T_{ref}$

→ delay induced phase margin degradation

- Rule of thumb: loop bandwidth $< 10\% f_{ref}$** limits PM degradation by 18°.
- The original loop transfer function does not even have T_{ref} as part of equation!!

$$H_{OL_ZOH}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)} \cdot \frac{1 - e^{-sT}}{sT}$$



Loop Dynamics Analysis

- Closed loop transfer function

$$H_{CL}(s) = \frac{H_{OL}(s)}{1 + H_{OL}(s)} = \frac{N(s)}{D(s)} = \frac{\frac{K_{PD} \cdot K_V}{RC}}{s^2 + \frac{s}{RC} + \frac{K_{PD} \cdot K_V}{RC}}$$

- Denominator is a second order system

$$D(s) = s^2 + 2\zeta\omega_n s + \omega_n^2, \omega_n = \sqrt{K_{PD} \cdot K_V / RC}, \zeta = 1 / (2\sqrt{K_{PD} \cdot K_V \cdot RC})$$

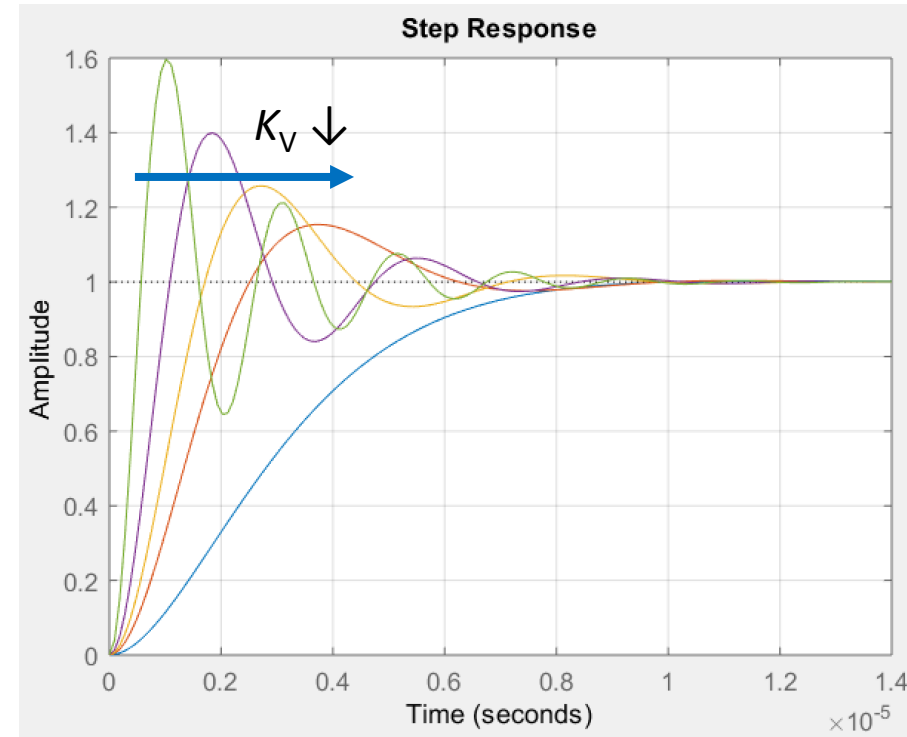
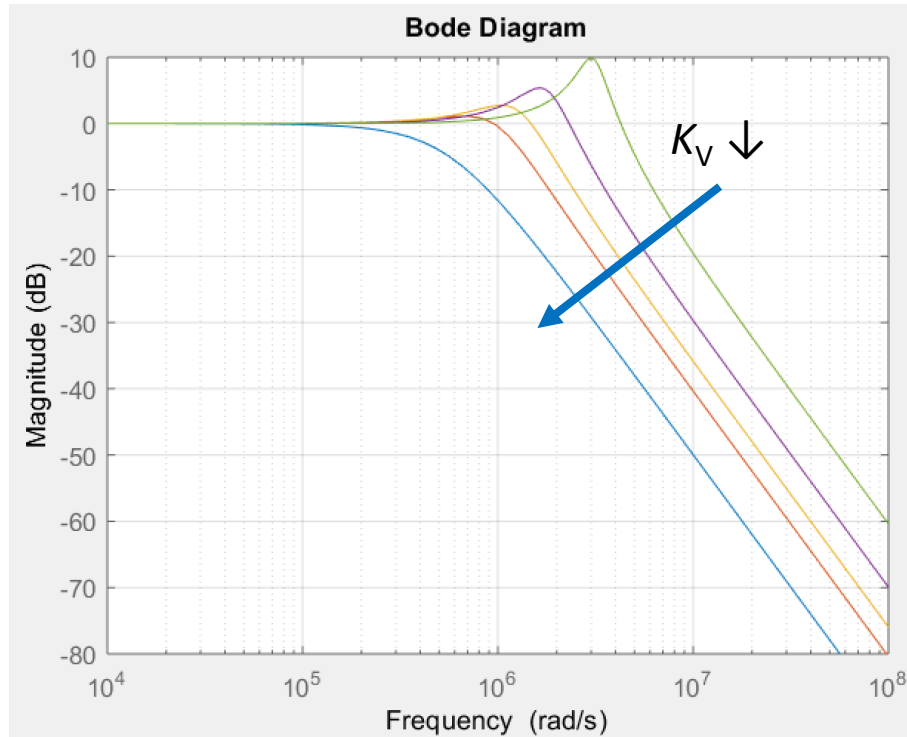
- Solving the second order system $D(s) = 0$

$$s = -\omega_n \left(\zeta \pm i\sqrt{1 - \zeta^2} \right), x(t) = e^{-\omega_n \zeta t} e^{i\omega_n \sqrt{1 - \zeta^2} t}$$

- $\zeta=0$: undamped, $x(t)$ oscillates at the natural frequency ω_n with a constant amplitude
- $0 < \zeta < 1$: underdamped, $x(t)$ is a decaying exponential $e^{-\omega_n \zeta t}$ with an oscillation frequency of $\omega_n \sqrt{1 - \zeta^2}$
- $\zeta > 1$: overdamped, $x(t)$ is a decaying exponential without oscillation
- $\zeta = 1$: critically damped is the border between the overdamped and underdamped cases.
- For PLL design, usually $\zeta \geq \sqrt{2}/2$ is desirable.



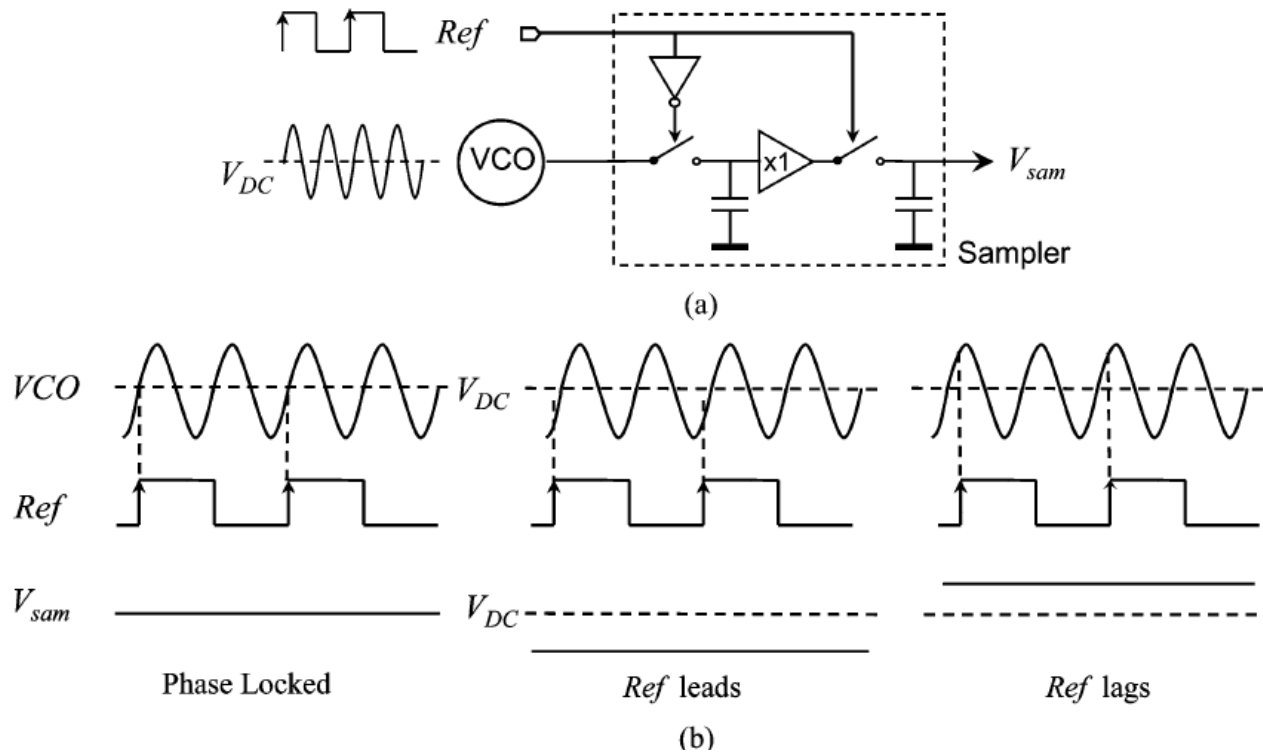
Loop Dynamics Analysis



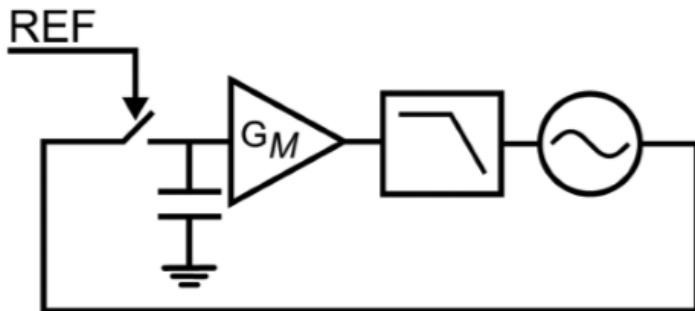
- Matlab commands for closed loop transfer function frequency response and transient step response
 - tf, bodemag, step



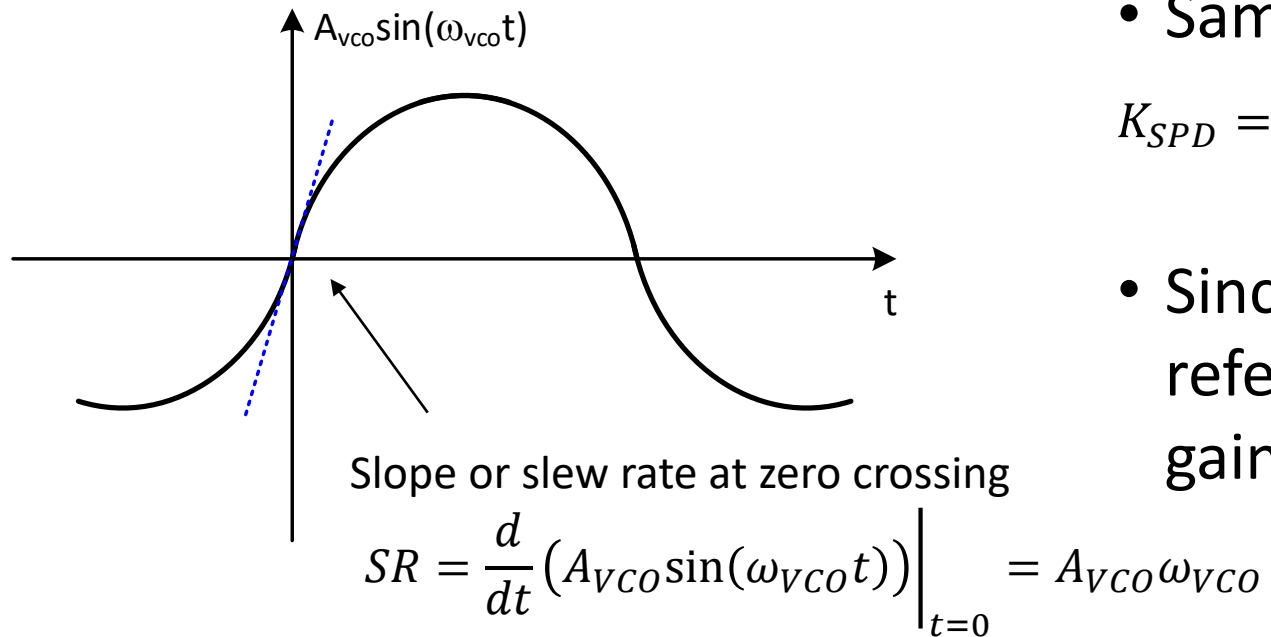
Subsampling Phase Detector and Subsampling PLLs



- Reference clock samples VCO output directly as phase detector
- Loop locks when f_{ref} rising edge aligns with VCO zero crossing
- High subsampling phase detector gain due to large VCO slew rate $\rightarrow$ suppress phase noise from following stages
- Integer-N PLL by nature. Use DTC to implement fractional-N functionality



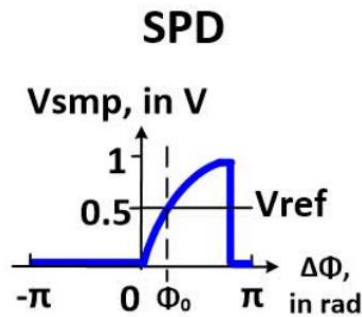
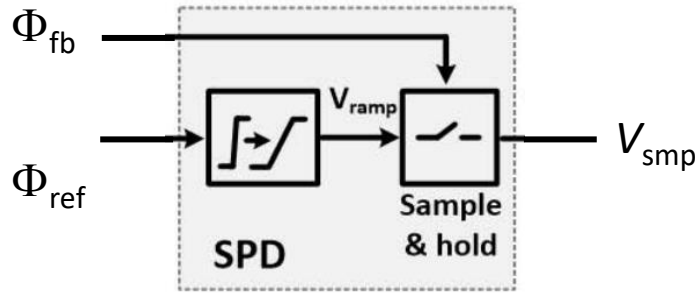
Subsampling Phase Detector Gain



- Sampling phase detector gain
$$K_{SPD} = \frac{dV}{dt} \frac{T_{ref}}{2\pi} = SR \frac{T_{ref}}{2\pi} = \frac{A_{VCO} \omega_{VCO} T_{ref}}{2\pi} = \frac{A_{VCO} f_{VCO}}{f_{ref}}$$
- Since ω_{VCO} is very large compared to reference frequency $1/T_{ref}$, subsampling PD gain is very high.



Sampling Phase Detector and Sampling PLLs



$$K_{SPD} = \frac{\Delta V}{\Delta \phi} = \frac{\Delta V}{\Delta T} \cdot \frac{\Delta T}{\Delta \phi} = K_{slope} \frac{T_{ref}}{2\pi} = \frac{K_{slope}}{2\pi f_{ref}}$$

Ref [6]

- Reference signal goes through a slope generator and becomes a ramping signal
- Feedback signal (VCO signal divided by N) samples the ramping reference
- The sampled voltage carries the timing or phase information.
- The steady state after loop locks, the sampled voltage equals to V_{ref}
- $K_{SPD} = \Delta V / \Delta \phi = K_{slope} / (2\pi f_{ref})$
- High sampling phase detector gain for large K_{SPD} → suppress phase noise from following stages
- Integer-N PLL by nature. Use DTC to implement fractional-N functionality



Phase Detector Types and PD Gain

Phase Detector Type	XOR	Mixer	Subsampling	Sampling
Input signals	Digital edges	Analog sinusoidal	RF sinusoidal & digital edge	Ramping signal & digital edge
Phase detector gain	$2V_0/\pi$	$\frac{\alpha A_1 A_2}{2}$	$\frac{A_{VCO} f_{VCO}}{f_{ref}}$	$\frac{K_{slope}}{2\pi f_{ref}}$

- XOR, mixer and subsampling phase detectors all have both positive and negative PD gain
- Subsampling and sampling phase detectors have high PD gain that suppress the phase noise contributions from the following stages
- Phase detector alone cannot detect frequency difference



Issues with Type-I PLL

- Limited acquisition range
 - When initial frequency difference between VCO and reference is large, PD output is periodic and has an average close to zero. The loop may never acquire lock.
 - PD gain is negative for part of input range that makes the loop positive feedback
- The loop stability (damping factor ζ and phase margin PM) is closely tied to the loop gain and loop bandwidth.
 - Lead to tradeoff between stability and loop bandwidth
 - Loop bandwidth needs to be independently optimized for phase noise, reference spurs, etc.
- Finite static phase error between reference and VCO that is VCO frequency dependent

$$\Delta\phi = \frac{\omega_{VCO} - \omega_0}{K_{PD} \cdot K_V}, \text{ where } \omega_{VCO} = \omega_0 + K_V \cdot V_{tune}$$



Type-I PLL S-domain Model Open Loop TF

```
% Type-I PLL
% Loop parameters
kpd=1/pi;
kv=[1e6 3e6 5e6 10e6 30e6];
r=1e3;
c=1e-9;
% Transfer function
for i=1:length(kv)
    num=[kpd*kv(i)];
    den=[r*c 1 0];
    H=tf(num, den)
    % Bode plot
    hold on;
    bode(H)
    grid on;
    [Gm(i),Pm(i),Wgm(i),Wpm(i)]=margin(H);
end;
figure(2);
yyaxis left;
plot(kv, Pm);
ylabel('Phase margin (deg)')
yyaxis right;
plot(kv, Wpm);
ylabel('f0dB (rad/s)')
xlabel('VCO gain (Hz/V)')
grid on;
```

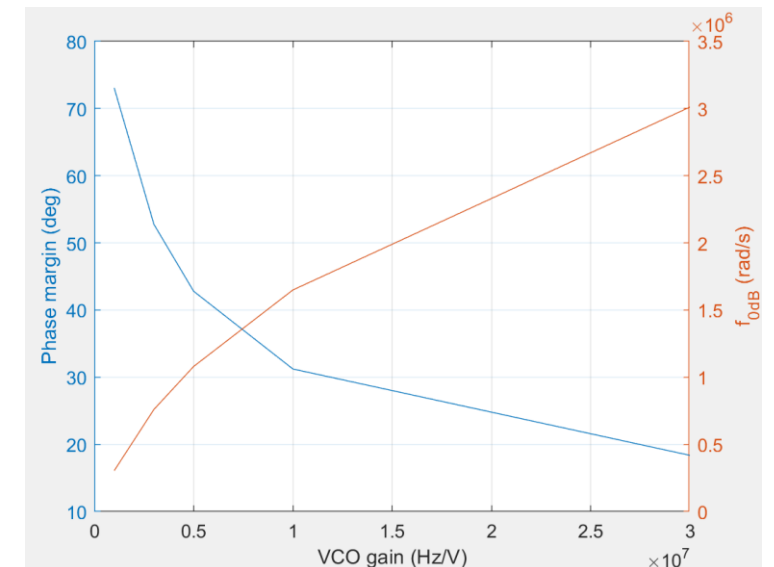
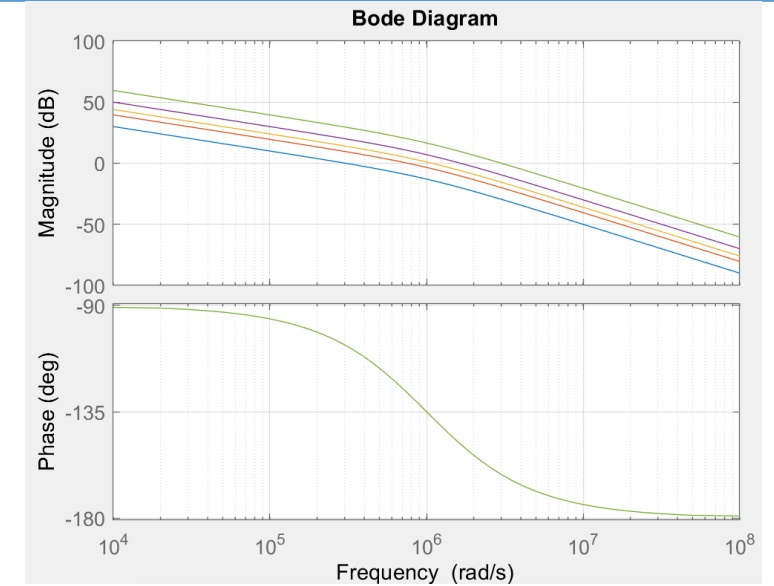
```
H =
    3.183e05
    -----
    1e-06 s^2 + s
Continuous-time transfer function.
```

```
H =
    9.549e05
    -----
    1e-06 s^2 + s
Continuous-time transfer function.
```

```
H =
    1.592e06
    -----
    1e-06 s^2 + s
Continuous-time transfer function.
```

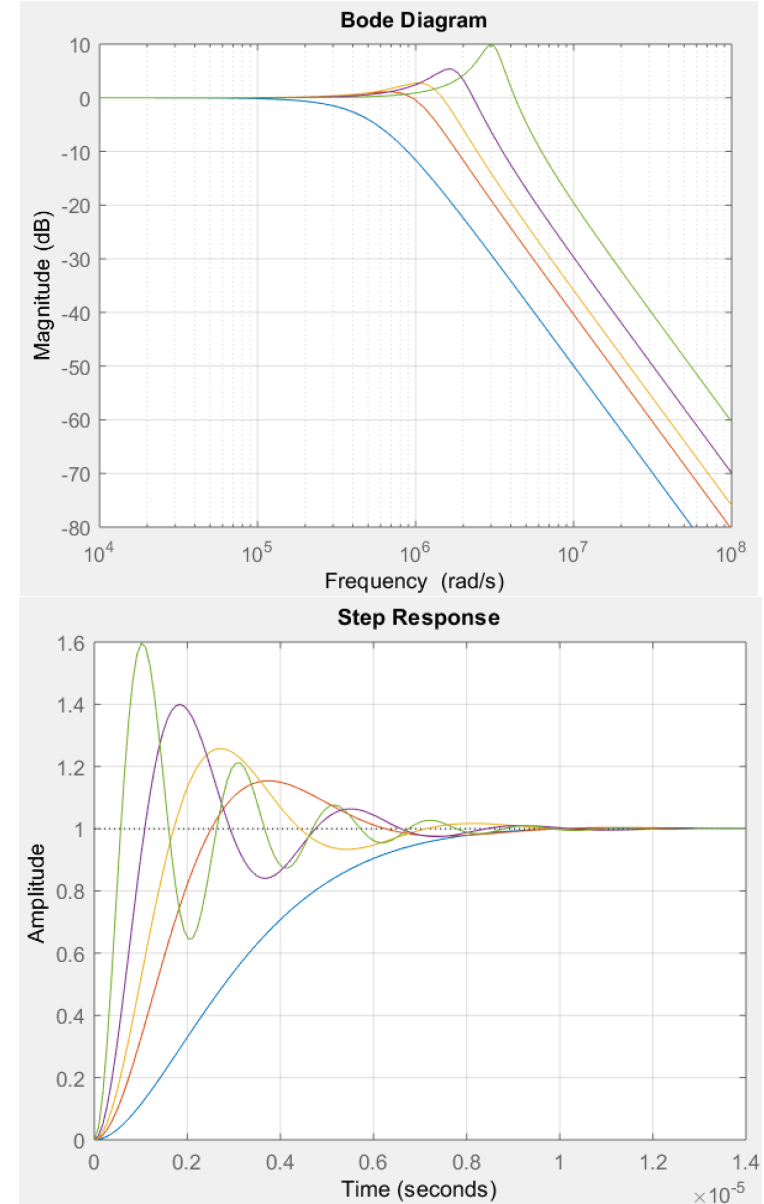
```
H =
    3.183e06
    -----
    1e-06 s^2 + s
Continuous-time transfer function.
```

```
H =
    9.549e06
    -----
    1e-06 s^2 + s
Continuous-time transfer function.
```



Type-I PLL S-domain Model Closed Loop TF

```
% Type-I PLL closed loop analysis
% Loop parameters
kpd=1/pi;
kv=[1e6 3e6 5e6 10e6 30e6];
r=1e3;
c=1e-9;
% Transfer function
for i=1:length(kv)
    num=[kpd*kv(i)/r/c];
    den=[1 1/r/c kpd*kv(i)/r/c];
    H_CL=tf(num, den);
    % Bode plot
    figure(1); hold on; bodemag(H_CL); grid on;
    % Step response
    figure(2); hold on; step(H_CL); grid on;
end;
```



References

- [1] Behzad Razavi, RF Microelectronics, Prentice Hall, 2nd edition, 2012
- [2] William F. Egan, Frequency Synthesis by Phase Lock, 2nd edition, 2000
- [3] X. Gao, E. Klumperink, M. Bohsali and B. Nauta, A 2.2GHz 7.6mW sub-sampling PLL with -126dBc/Hz in-band phase noise and 0.15psrms jitter in 0.18um CMOS, pp 392-393, ISSCC 2009
- [4] N. Markulic, et al., A self-calibrated 10Mb/s phase modulator with -37.4dBc EVM based on a 10.1-to-12.4GHz - 246.6dB-FOM, fractional-N subsampling PLL, pp 176-177, ISSCC 2016
- [5] X. Gao, E. Klumperink, M. Bohsali and B. Nauta, A low noise sub-sampling PLL in which divider noise is eliminated and PFD/CP noise is not multiplied by N^2 , JSSC 2009
- [6] W. Wu, et al., A 28-nm 75-fsrms Analog Fractional-N Sampling PLL With a Highly Linear DTC Incorporating Background DTC Gain Calibration and Reference Clock Duty Cycle Correction, JSSC 2019

